

i Examination paper for IT3708 – Bio-inspired Artificial Intelligence

Date: May 13, 2025

Time: 09:00

Course contact: Xavier F. C. Sánchez Díaz

Present at the exam location: NO

Permitted examination support material: D / No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment.

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website “Explanation of grades and appeals”.

SPECIFIC INFORMATION FOR YOUR COURSE

No paper drawings:

This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. You will not be able to submit the sheets, and they will not be graded.

Weighting:

Each “multiple choice” question has exactly one answer and is worth 1 point, for a total of 16 points. The “written answer” questions are worth a total of 14 points. For each “written answer” question, the maximum score is given with the question. The total number of points for the exam is 30.

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspera. Be aware that it may take a working day until any hand-written material is available in “previous tests”.

- 1 Among the following statements, pick one **correct statement** about the role of diversity maintenance in multi-objective optimization using genetic algorithms.

Select one alternative:

- ☐ Diversity maintenance in multi-objective genetic algorithms is unnecessary, as the algorithm naturally produces evenly distributed solutions across the entire Pareto front without any specialized techniques.
- ☐ Once a set of non-dominated solutions is found, diversity maintenance can be discontinued, as the algorithm will automatically preserve a well-distributed Pareto front throughout subsequent generations.
- ☐ Maintaining diversity in multi-objective optimization is achieved most effectively by repeatedly selecting identical individuals from the population to reinforce convergence toward a single optimal solution.
- ☐ Diversity maintenance in multi-objective genetic algorithms is crucial for ensuring a well-distributed Pareto front, enabling the algorithm to explore trade-offs effectively and avoid premature convergence to suboptimal regions.
- ☐ Diversity mechanisms in multi-objective genetic algorithms are only applicable in single-objective problems and are considered counterproductive when handling more than one objective function.

Maximum marks: 1

2 Among the following statements, pick one **correct statement** about the role of multi-objective optimization in bio-inspired algorithms.

Select one alternative:

- ☐ Bio-inspired algorithms do not require multi-objective optimization techniques because natural systems inherently evolve toward only one optimal solution in all environmental contexts.
- ☐ Multi-objective optimization plays a pivotal role in bio-inspired algorithms by enabling the simultaneous exploration of trade-offs among conflicting objectives, often yielding a diverse set of Pareto-optimal solutions.
- ☐ All bio-inspired algorithms that apply multi-objective optimization techniques are guaranteed to converge to the true Pareto front within a fixed number of generations, regardless of problem complexity.
- ☐ In bio-inspired algorithms, multi-objective optimization operates under the assumption that all objectives are linearly dependent, thereby simplifying the problem into a single-objective model.
- ☐ The role of multi-objective optimization in bio-inspired algorithms centers on decorative visualizations of trade-off curves and holds no practical significance in real-world engineering applications.

Maximum marks: 1

- 3 Among the following statements, pick one **correct statement** about the role of mutation and crossover in multi-objective genetic algorithms.

Select one alternative:

- ☐ In multi-objective genetic algorithms, the crossover operator is primarily responsible for evaluating objective functions and does not contribute to the generation of new candidate solutions.
- ☐ In multi-objective genetic algorithms, the mutation operator acts as a selection mechanism, filtering out infeasible individuals rather than introducing variation into the population.
- ☐ In multi-objective genetic algorithms, mutation and crossover are fundamental variation operators that generate diverse offspring and facilitate exploration of the solution space to approximate the Pareto-optimal front.
- ☐ The use of both crossover and mutation in multi-objective genetic algorithms is discouraged, as the operators disrupt the non-dominated sorting process and introduce inconsistencies in Pareto front estimation.
- ☐ Mutation in the NSGA-II multi-objective genetic algorithm is deterministic and designed to move solutions directly toward the center of the objective space, ignoring diversity preservation.

Maximum marks: 1

- 4 Among the following statements, pick one **correct statement**. The statements are concerned about the use of Markov chains to analyze genetic algorithms.

Select one alternative:

- ☐ The ease of using Markov chains to analyze genetic algorithms derives from the assumption that populations consist of independent individuals, which eliminates interactions and simplifies modeling efforts.
- ☐ Because genetic algorithms operate deterministically once initialized, Markov chain analysis can reliably predict exact future populations without consideration of random variation introduced by crossover or mutation.
- ☐ Genetic algorithms inherently induce small and manageable state spaces, enabling straightforward construction of transition matrices and exact calculation of steady-state distributions via standard Markov chain modeling.
- ☐ Analyzing genetic algorithms by means of Markov chains is in principle possible but in practice difficult, due to large state spaces and stochastic transitions.
- ☐ Due to the fact that genetic algorithms are designed primarily for empirical applications, formal analysis through Markov chains is impossible and offers no theoretical insights.

Maximum marks: 1

- 5 Among the following statements about the role of computational complexity theory in bio-inspired algorithms, which one is **correct**?

Select one alternative:

- ☐ There is no connection between computational complexity theory and bio-inspired algorithms, as the former deals only with theoretical models while the latter operates entirely outside of formal analysis including computational complexity theory.
- ☐ Computational complexity theory is not relevant to bio-inspired algorithms, because their stochastic nature allows them to solve any problem, including undecidable ones, without concern for time or resource constraints.
- ☐ Computational complexity theory helps in understanding the limitations of bio-inspired algorithms, especially for NP-hard problems, where they offer approximate or heuristic solutions without guarantees of optimality or polynomial runtime.
- ☐ Complexity theory discourages the use of bio-inspired algorithms, as it classifies all heuristic and metaheuristic methods as inefficient and fundamentally unprovable in terms of performance or scalability.
- ☐ Bio-inspired algorithms are not affected by NP-hardness since they use randomization, which transforms any hard problem into an easy one through iteration over a number of generations with an arbitrarily sized population, thus mimicking natural evolution.

Maximum marks: 1

6 What are the pros and cons of using genetic algorithms to solve NP-complete problems? Among the following statements, which statement is **correct**?

Select one alternative:

- ☐ Genetic algorithms provide a useful heuristic approach for approximating solutions to NP-complete problems, though they don't guarantee optimality and can be sensitive to their parameters and problem encoding.
- ☐ As genetic algorithms mimic natural evolution, they typically avoid local minima and guarantee finding global optima in problems of practical interest, regardless of their computational complexity or dimensionality.
- ☐ Since genetic algorithms can be formalized by means of Markov chains informed by the algorithms' operators and parameters, these algorithms are not impacted by the NP-hardness results of computational complexity theory.
- ☐ The performance of a genetic algorithm is strongly affected by the encoding of the problem it is solving, and as a consequence computational complexity depends on the algorithm's encoding choice, rather than the nature of the problem itself.
- ☐ Using genetic algorithms is always more efficient than traditional heuristics, as they require minimal parameter tuning and perform close to optimally with default crossover and mutation rates, making them suitable to any NP-complete problem.

Maximum marks: 1

- 7 The concept of neighborhood is crucial to define a search landscape. Which of the following statements about the neighborhood is **false**?

Select one alternative:

- ☐ The neighborhood of a solution x is the set of solutions that are within a given distance of x .
- ☐ The neighborhood of a bitstring b is usually considered as the set of bitstrings that can be reached with one bitflip.
- ☐ A bitstring of size n has up to n neighbors when we define the neighborhood as having a Hamming distance of size 1.
- ☐ Two bitstrings have the same neighborhood if they have the same number of 1s.
- ☐ The neighborhood in a permutation vector is, for example, the set of all permutations that can be reached in one mutation step.

Maximum marks: 1

- 8 Particle Swarm Optimization (PSO) is a metaheuristic inspired by bird flocks. Which of the following statements about PSO is **false**?

Select one alternative:

- ☐ Particles in PSO consider the positions and velocities of their neighbors.
- ☐ In PSO, each particle has a position, a velocity, and keeps a record of its personal best position.
- ☐ PSO was proposed as a metaheuristic for continuous optimization.
- ☐ In PSO, the size of a particle's neighborhood will define how many particles will influence that particle's movement.
- ☐ Each particle in PSO adds pheromone to a trail, making it more likely to be followed by other particles in the swarm.

Maximum marks: 1

- 9 In constrained optimization, the final solution needs to be feasible. Which of the following statements is **true** about feasibility?

Select one alternative:

- ☐ Searching only in the feasible areas of a search landscape will not ensure reaching the global optimum.
- ☐ Starting with a 100% feasible population will ensure that the population of the next generation is 100% feasible.
- ☐ A feasible solution will always have a better fitness than an infeasible solution since the infeasible solution will be heavily penalized.
- ☐ Performing crossover of two feasible solutions will ensure that the offspring are feasible solutions.
- ☐ Performing mutation on an infeasible solution will ensure feasibility if the mutation probability parameter is high enough.

Maximum marks: 1

- 10 Throughout the course, we studied several problems; both discrete (or combinatorial) and continuous. Which of the following problems is **not** combinatorial?

Select one alternative:

- ☐ Vehicle Routing Problem with time windows
- ☐ Binary Knapsack Problem
- ☐ The n -dimensional Ackley Function
- ☐ Travelling Salesperson Problem
- ☐ Feature Selection

Maximum marks: 1

11 Which of the following statements is **not** a way to escape local optima?

Select one alternative:

- ☐ Gradually reduce the frequency of taking a non-optimal decision
- ☐ Use elitism to increase the selective pressure
- ☐ Restart the search process
- ☐ Make non-optimal decisions every so often
- ☐ Apply perturbations to promising solutions

Maximum marks: 1

12 During the course, we studied search landscapes to understand how our algorithms traverse the search space. Which of the following statements about landscape analysis is **false**?

Select one alternative:

- ☐ A search landscape depends on a fitness function, a search space, and a neighborhood definition
- ☐ A search landscape can be generated by running an algorithm until it finds all local optima in the search space
- ☐ Fitness landscapes were originally proposed in the study of Genetics (to explain why some species stopped evolving)
- ☐ Different algorithms will create different search landscapes of the same fitness function
- ☐ Studying search landscapes can help assessing problem difficulty and explain algorithm behavior

Maximum marks: 1

- 13 As in solvers in machine learning, many of the metaheuristics and algorithms in evolutionary computation need some sort of parameter setting. Which of the following statements regarding parameters is **false**?

Select one alternative:

- ☐ Encoding mutation rates as part of an individual's chromosome is a form of self-adaptive parameter control.
- ☐ Increasing the mutation rate after every k successful mutations in Evolutionary Strategies is a form of self-adaptive parameter control.
- ☐ Parameter tuning is performed by the user, and it relies on the user's knowledge about the metaheuristic and about the problem being solved.
- ☐ Setting a rule for increasing the mutation rate every k generations is a form of deterministic parameter control.
- ☐ Parameters can be set up and adjusted before the run and during the run, which we call parameter tuning and parameter control, respectively.

Maximum marks: 1

- 14 Which of the following statements is **not true** about the single point crossover (1PX) when applied in constrained optimization?

Select one alternative:

- ☐ 1PX can never keep together genes from opposite ends of the string.
- ☐ By using 1PX, genes that are near each other are more likely to be kept together.
- ☐ 1PX is an exploitation technique.
- ☐ Using 1PX on two feasible individuals (when dealing with permutation vectors) ensures that the crossover results in at least one feasible solution.
- ☐ 1PX works for both continuous and binary representations in constrained optimization.

Maximum marks: 1

- 15 Evaluate the following statements about the population management in an evolutionary algorithm and pick the one that is **true**.

Select one alternative:

- ☐ How selection and replacement work together does not need to be considered.
- ☐ The choice of a selection method depends entirely on the crossover method used.
- ☐ Selection in evolutionary algorithms is often performed by means of local search.
- ☐ Selection from the population does not depend on the chromosome representation.
- ☐ When you perform selection in a genetic algorithm, there is no need to do replacement.

Maximum marks: 1

- 16 Memetic Algorithms (MA) are hybrid algorithms that integrate an Evolutionary Algorithm and some other algorithm, for example Stochastic Local Search. Which of the following statements is **not** true about MAs?

Select one alternative:

- ☐ When including known good solutions, average performance is often increased. However, best performance can come from a randomized inclusion of solutions
- ☐ Initializing populations with already known solutions is a common practice in MAs
- ☐ Since Local Search considers neighborhoods, it can only be applied to permutation problems
- ☐ Constructive heuristics are commonly used, for example "Maximum number of conflicts first" in Constraint Optimization Problems
- ☐ To incorporate good solutions into the population, Stochastic Local Search can be used to climb from random individuals in the population

Maximum marks: 1

- 17 Some problems are very computationally expensive, like the Home-Care Vehicle Routing Problem where you had to assign nurses to routes and patients.

Describe at least **three ways** of speeding up the performance of an evolutionary algorithm in cases such as this, and use **no more than 4 sentences** for each way.

Fill in your answer here

Format	▼					↺			
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									Words: 0

Maximum marks: 6

- 18** Imagine that you are, as an expert on AI including bio-inspired AI, advising a client interested in implementing software to approximately solve a hard computational problem. The client recently came across a proof that the problem is NP-complete, and is asking you this: *What are the implications of the fact that the problem is NP-complete?*

Briefly discuss **three possible answers** to the question, and justify each answer with a few sentences. Each justification should be **at most 4 sentences long**. You can assume that P is not equal to NP, and you can further assume that the NP-completeness result for the problem in question is correct.

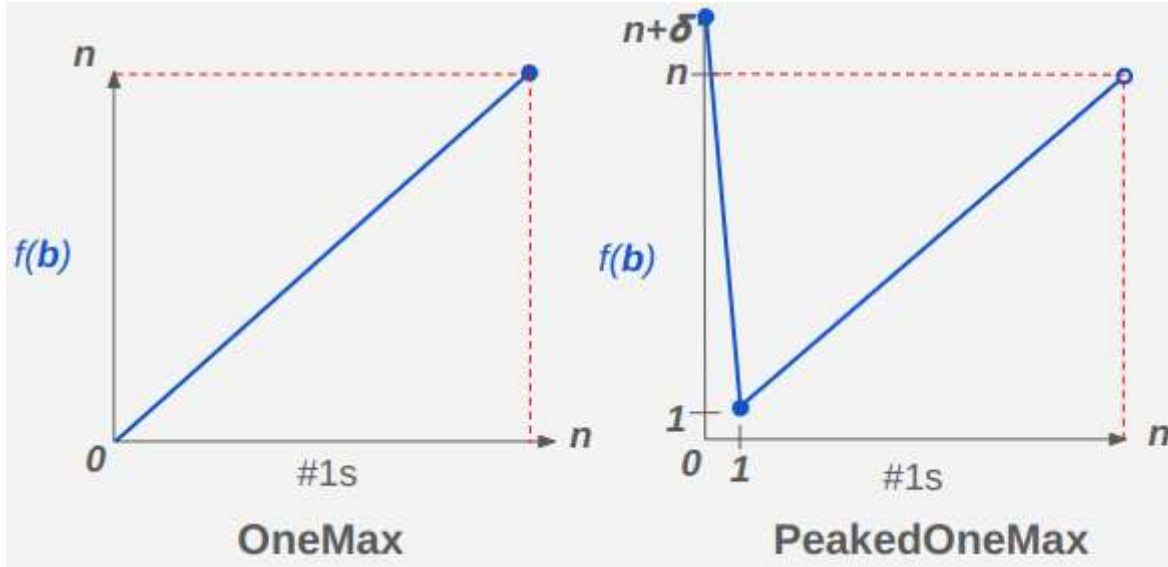
Fill in your answer here

Maximum marks: 6

19 **OneMax** is a function in which the fitness $f(\mathbf{b})$ is the sum of 1s in a bitstring $\mathbf{b}$. The maximum is achieved when all n bits are set to 1.

PeakedOneMax is a deceptive function that has its maximum when all n bits are set to 0, with $f(\mathbf{0}) = n + \delta$, for any $0 < \delta < n$. Otherwise, this function behaves like **OneMax**.

Here is a figure with the **OneMax** problem on the left, and the **PeakedOneMax** problem on the right. In both cases, the fitness $f(\mathbf{b})$ is plotted as a function of the number of bits set to 1 in a bitstring $\mathbf{b}$.



How could we show that one of the functions is harder to solve than the other, for a given bio-inspired algorithm $\mathcal{A}$? Describe briefly (in **no more than 4 sentences**) at least **1 method** for doing so (either analytical or experimental).

Fill in your answer here

Words: 0

Maximum marks: 2